

AYE
TECH HUB

• MECHANICAL ENGINEERING • LESSON 04

Thermodynamics

Basics

Lesson 04 of 30 | Mechanical Engineering Program

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- The four laws of thermodynamics — explained simply
- Heat, work, internal energy and enthalpy defined
- The Carnot cycle and why no engine can be 100% efficient
- Three heat transfer modes: conduction, convection, radiation
- Thermodynamics in engines, refrigerators and power plants

Thermodynamics — The Science of Energy

Thermodynamics is the branch of physics and engineering that deals with **heat, work, temperature, and energy transfer**. Every mechanical system that generates, converts, or uses energy is governed by thermodynamic principles — from the engine in a car to the turbine in a power station, from a refrigerator to the human body. The word comes from Greek: *therme* (heat) + *dynamis* (power). Mastering thermodynamics is non-negotiable for mechanical engineers working on engines, HVAC, power generation, turbines, and any system where energy changes form.

The Four Laws of Thermodynamics

ZEROth LAW	$T_A = T_B = T_C$	If A is in thermal equilibrium with B, and B with C, then A is in equilibrium with C. <i>Defines temperature. Basis for thermometers and temperature measurement.</i>
FIRST LAW	$\Delta U = Q - W$	Energy cannot be created or destroyed — only converted from one form to another. <i>Internal energy change = heat added – work done by system.</i>
SECOND LAW	$\Delta S_{\text{universe}} \geq 0$	Entropy of an isolated system always increases. Heat flows naturally from hot to cold. <i>Why engines are < 100% efficient. Why time has a direction.</i>
THIRD LAW	$S \rightarrow 0 \text{ as } T \rightarrow 0 \text{ K}$	As temperature approaches absolute zero (0 K), entropy approaches a minimum. <i>Sets the absolute reference for entropy. No process can reach 0 K.</i>

Figure 1 — The Four Laws of Thermodynamics (listed from Third to Zeroth)

Concept	Symb ol	Definition	Unit
Temperature	T	Average kinetic energy of molecules. Kelvin scale starts at absolute zero.	K (Kelvin) or °C
Heat	Q	Thermal energy transferred between systems due to a temperature difference.	Joule (J) or kJ
Work	W	Energy transferred by force acting through a distance. $W = \int P \, dV$.	Joule (J) or kJ
Internal Energy	U	Total kinetic + potential energy of all molecules in a system.	Joule (J) or kJ
Enthalpy	H	$H = U + PV$. Convenient for flow processes (turbines, compressors, nozzles).	Joule (J) or kJ
Entropy	S	Measure of molecular disorder. Increases with irreversibility. $\Delta S = Q/T$.	J/K or kJ/K
Specific Heat	c_p, c_v	Energy needed to raise 1 kg of substance by 1 K at constant P or V.	J/(kg·K)

First Law simplified: $\Delta U = Q - W$. If you add 100 J of heat to a gas ($Q = +100 \text{ J}$) and the gas does 40 J of work expanding against a piston ($W = +40 \text{ J}$), the internal energy increases by 60 J. The sign convention: Q positive when heat enters the system; W positive when work is done BY the system.

Heat Transfer — How Energy Moves

The Second Law tells us heat flows naturally from hot to cold. But *how* it flows depends on the medium. There are exactly three modes — and every real heat transfer problem involves one, two, or all three simultaneously.

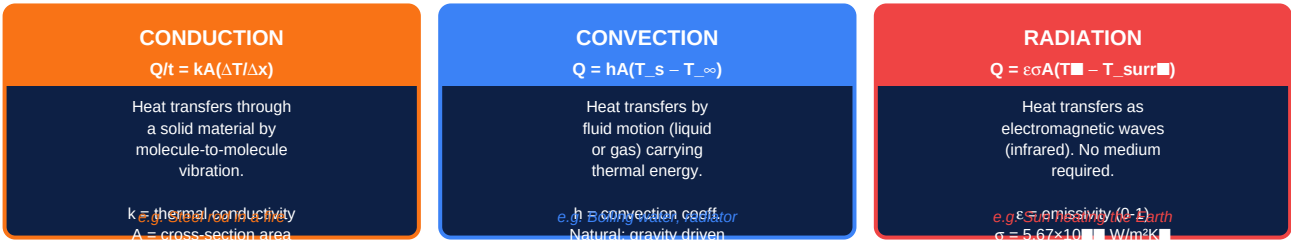


Figure 2 — Three Modes of Heat Transfer with Governing Equations

Mode	Medium Required	Rate Equation	Typical Application
Conduction	Solid (or stationary fluid)	$q = -kA(dT/dx)$ Fourier's Law	Heat loss through a wall, fin heat sinks, cooking pot
Natural Convection	Fluid — gravity-driven buoyancy flow	$q = hA(T_s - T_\infty)$ $h = 2\text{--}25 \text{ W/m}^2\text{K}$ (air)	Room heating from a radiator, hot pipe in still air
Forced Convection	Fluid — flow driven by fan, pump, wind	$q = hA(T_s - T_\infty)$ $h = 25\text{--}250 \text{ W/m}^2\text{K}$ (air)	Air-cooled heat sink with fan, water cooling, car radiator
Radiation	None — electromagnetic waves (infrared)	$q = \epsilon\sigma A(T_s^4 - T_{surr}^4)$ Stefan-Boltzmann Law	Solar heating, furnace walls, spacecraft thermal control

Thermal Conductivity (k)	Material	k Value (W/m·K)
Very high (metals)	Copper	385 Aluminium: 205 Steel: 50
Medium	Concrete / Glass	1.0–1.5 Water: 0.6
Low (insulators)	Wood / Brick	0.1–0.15 Fibreglass: 0.04
Very low (insulation)	Aerogel / PIR foam	0.013–0.022 (excellent insulator)
Near zero	Air (still)	0.025 Argon: 0.018

Thermal resistance analogy: Heat conduction through a wall is analogous to electrical current through a resistor. $R_{\text{thermal}} = L / (k \times A)$. Multiple layers in series add resistances: $R_{\text{total}} = R_{\text{brick}} + R_{\text{insulation}} + R_{\text{plaster}}$. This is how U-values for building elements are calculated.

The Carnot Cycle and Real Engine Efficiency

The **Carnot cycle** (Sadi Carnot, 1824) is a theoretical ideal heat engine cycle that defines the maximum possible efficiency between two temperature reservoirs. No real engine can exceed Carnot efficiency — it is the absolute upper bound set by the Second Law.

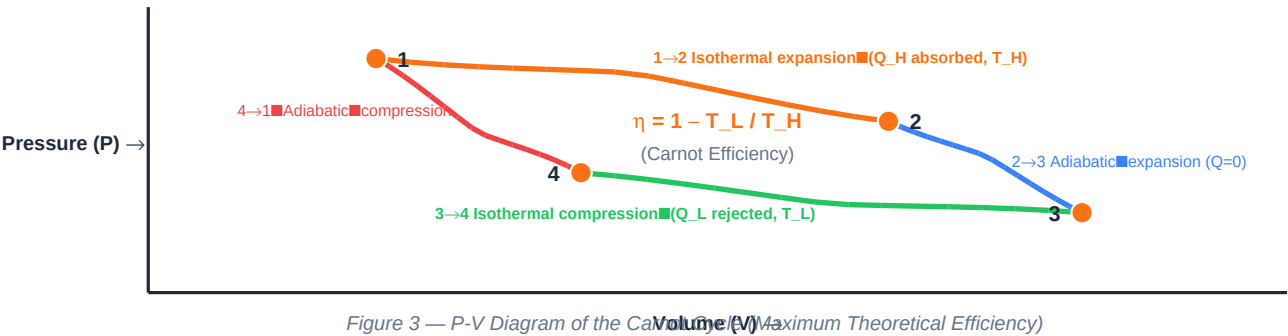


Figure 3 — P-V Diagram of the Carnot Cycle (Maximum Theoretical Efficiency)

Engine Type	Typical Efficiency	Hot Source (T _H)	Cold Sink (T _L)	Carnot Limit
Petrol engine	25–35%	~2500°C combustion (780K avg cycle)	~60°C exhaust (333K)	$\eta_{\text{max}} = 1 - 333/780 = 57\%$
Diesel engine	40–45%	~1000°C peak (600K avg)	~80°C exhaust (353K)	$\eta_{\text{max}} = 1 - 353/600 = 41\%$
Steam turbine (power plant)	35–42%	~550°C steam (823K)	~30°C condenser (303K)	$\eta_{\text{max}} = 1 - 303/823 = 63\%$
Gas turbine (jet engine)	35–45%	~1400°C (1673K)	~15°C ambient (288K)	$\eta_{\text{max}} = 1 - 288/1673 = 83\%$
Combined cycle gas turbine	55–62%	~1400°C (1673K)	~30°C condenser (303K)	$\eta_{\text{max}} = 83\%$; real loss reduced by heat recovery

Thermodynamic Cycles and Applications

Cycle	Processes	Application	Key Formula
Carnot	Isothermal expansion → adiabatic expansion → isothermal compression → adiabatic compression	Theoretical maximum; not used in practice due to slow isothermal processes	$\eta = 1 - T_L/T_H$
Otto Cycle	Adiabatic compression → isochoric heat addition → adiabatic expansion → isochoric heat rejection	Petrol/gasoline engine (spark ignition)	$\eta = 1 - 1/r^{(\gamma-1)}$ r = compression ratio
Diesel Cycle	Adiabatic compression → isobaric heat addition → adiabatic expansion → isochoric rejection	Diesel engine (compression ignition)	Higher r → higher efficiency than Otto
Rankine Cycle	Pump → boiler → turbine → condenser	Steam power plants, nuclear power, geothermal	$\eta = W_{\text{net}} / Q_{\text{in}}$; uses phase change (liquid → steam)
Reversed Rankine (Refrigeration)	Compressor → condenser → expansion valve → evaporator	AC systems, refrigerators, heat pumps — covered in HVAC program	$\text{COP} = Q_L / W_{\text{in}}$

Brayton Cycle	Compressor → combustion chamber → turbine → heat rejection	Gas turbines, jet engines, industrial gas turbines	$\eta = 1 - 1/r_p^{(\gamma-1)/\gamma}$ r_p = pressure ratio
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What comes next — MECH-005: Fluid Mechanics — fluid properties, pressure, Bernoulli's equation, pipe flow, Reynolds number, and pump selection. The foundation for designing any system that moves liquid or gas.